Entropy as a Projection Artifact: One-Body Operators are Entanglement-Inert on Sparse Lattices

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We characterize the operator-theoretic origin of entanglement entropy in the GeoVac Coulomb lattice, a finite spectral graph realization of the hydrogenic angular momentum eigenbasis on S^3. Five results are presented. First, the von Neumann entropy of any single-particle reduced density matrix derived from a *non-degenerate* ground state of a purely one-body fermionic Hamiltonian on a finite lattice is identically zero; numerically, we confirm $S_{\mathrm{kin}}/S_{\mathrm{full}} \sim 10^{-14}$ for He at $n_{\mathrm{max}} = 2, 3$ in the angular momentum eigenbasis. All ground-state entanglement is generated by the two-body interaction V_{ee}, with the non-degeneracy qualifier essential: open-shell and dissociated-bond states escape the floor. Second, the holographic area-law scaling $S_n = k \log A_n$ with $A_n \propto n^4$, asserted on one-particle grounds in earlier framework papers, is rigorously recovered as a pair-counting statement: $A_n = g_n^2$, where $g_n = 2n^2$ is the shell degeneracy. The factor of four is the two-body signature. Third, in the Coulomb lattice the cusp concentrates as a single "hot node" on the $(1s, 1s)$ pair-state of the V_{ee} graph; entanglement is co-located with cusp mass. Fourth, on the Bargmann–Segal harmonic oscillator lattice the ground-state entropy of any two-fermion closed-shell state interacting via *any* central two-body potential is identically zero, a consequence of Moshinsky–Talmi total-quanta conservation; this is a rigidity statement structurally parallel to the Fock and Bargmann–Segal theorems of Paper 24. Fifth, Coulomb-lattice block entropy obeys a universal scaling $S_B = A\,(\tilde{w}_B/\delta_B)^{\gamma}$, where $\tilde{w}_B$ is the dimensionless V_{ee} mass off-diagonal in the one-body eigenbasis and δ_B the dimensionless first H_1 gap; the local exponent extrapolates to $\gamma_\infty \approx 1.96$ in the joint $Z, n_{\mathrm{max}} \to \infty$ limit (Richardson on $n_{\mathrm{max}} = 2, 3, 4, 5$), slightly below the non-degenerate second-order Rayleigh–Schrödinger value of 2, with the residual gap attributed to multi-shell aggregation in the basis-extension limit. Entropy is slotted into the exchange-constant taxonomy as a projection-class observable: zero on the combinatorial graph, finite only after the metric embedding of $1/r_{12}$.

I. INTRODUCTION

The kinetic and potential parts of a fermionic Hamiltonian play asymmetric roles in generating entanglement. Slater determinants exactly diagonalize one-body operators and carry only the entanglement floor mandated by antisymmetrization; all non-trivial orbital correlation arises from two-body terms. This fact is folklore in the quantum-chemistry and quantum-information literatures (see, e.g., [1, 2]), but its sharp form is rarely written out, and its consequences on *sparse, lattice-realized* Hamiltonians are seldom exhibited explicitly.

The GeoVac framework [4, 6, 7] provides a controlled testbed: a finite spectral graph on the angular momentum eigenbasis of S^3, where the one-body operator H_1 is the graph Laplacian (with universal coupling $\kappa = -1/16$) plus the diagonal hydrogenic spectrum, and the two-body operator V_{ee} is the Coulomb repulsion $1/r_{12}$ projected onto the same vertex set. Paper 26 established that, in this basis, the off-diagonal one-body Hamiltonian carries 10–39% of the correlation energy but $< 0.2\%$ of the single-orbital entanglement entropy across He-like ions [11]. Paper 26 leaves open whether this asymmetry is a quantitative numerical observation specific to the basis at hand or a structural identity that holds at the operator level.

This paper resolves the question. Section II states the operator-theoretic fact: on any finite lattice, the one-body operator is entanglement-inert at the closed-shell ground state. Section III confirms the result to machine precision on He at $n_{\mathrm{max}} = 2, 3$. Section IV re-derives the holographic area-law $S_n \propto 4 \log n$ as a pair-counting statement, fixing a mislabeling in Paper 5 [5] where the scaling was presented in one-particle language. Section V shows that the cusp localizes as a single hot node on the two-body V_{ee} graph, making entropy and cusp mass spatially co-located. Section VI places entropy in the exchange-constant taxonomy of Paper 18 [8] as a projection-class observable. Section VII states two falsifiable predictions, one for the Bargmann–Segal harmonic oscillator lattice [10] and one across the GeoVac molecular library.

Provenance (register tiers). The entanglement-inertness of the one-body operator (§ II) and the harmonic-oscillator zero-entropy rigidity (§ VII) are INTERNAL THEOREMS; the $S_{\mathrm{kin}}/S_{\mathrm{full}} \sim 10^{-14}$ numerical confirmation and the block-entropy exponent $\gamma_\infty \approx 1.96$ (a Richardson extrapolation, *not* an exact value) are MEASURED; the area-law $A_n = g_n^2$ recovery is an exact combinatorial identity (SYMBOLIC). Entropy is placed as an embedding-class OBSERVATION in the Paper 18 exchange-constant taxonomy.

II. OPERATOR-THEORETIC RESULT

A. Setting

Let $\mathcal{B} = \{\phi_1, \ldots, \phi_M\}$ be a finite orthonormal single-particle basis on a lattice (the vertex set of a spectral graph), and let

$$H_1 \;=\; \sum_{ij} h_{ij}\, c_i^\dagger c_j, \qquad h = h^\dagger, \tag{1}$$

be a one-body Hermitian operator on the antisymmetric Fock space generated by $\mathcal{B}$. Let Φ_0 be the ground state of H_1 at fixed particle number N.

B. Statement

Proposition. *If the spectrum of h is non-degenerate at the Fermi level (i.e., Φ_0 corresponds to a uniquely defined Slater determinant of the N lowest natural orbitals of h), then for any subset $A \subset \mathcal{B}$ of orbitals the von Neumann entropy*

$$S_A \;=\; -\mathrm{Tr}\left[\rho_A \log \rho_A\right], \qquad \rho_A \;=\; \mathrm{Tr}_{\mathcal{B}\setminus A}\, |\Phi_0\rangle\langle\Phi_0|, \tag{2}$$

is determined entirely by the natural-orbital occupation spectrum of Φ_0 restricted to A, which consists of zeros and ones in the natural-orbital basis. Equivalently, in the basis that diagonalizes h, ρ_A is a tensor product of single-mode density matrices each of which has support on a single occupation eigenvalue. In a generic basis $\mathcal{B}$, the resulting entropy is the antisymmetrization (Slater) entropy [3], and is independent of h.

The content of the proposition is well known in quantum-information formulations of free-fermion systems [1]: a Slater determinant has a Gaussian correlation matrix, and its reduced density matrices are fully characterized by the projector $C = P_A^\dagger P P_A$ onto the occupied subspace, restricted to the partition A. The eigenvalues of C are pinned to $\{0, 1\}$ in the natural basis; any positive entropy of ρ_A in a different basis is a basis-projection artifact, not a physical correlation.

C. Scope of the proposition

The non-degeneracy clause is essential. An empirical scan (EP-2j) at $n_{\max} = 3$ in the GeoVac S^3 basis with $N = 3, 4$ electrons shows two distinct violation modes:

Open-shell single Slater determinant (Li, ^2S ground state). The H_1 ground state is non-degenerate (gap $\sim 10^{-5}$ Ha) and is exactly one Slater determinant $|1s^2\, 2s\rangle$, so the proposition's premise holds in the canonical statement of Sec II. However, the spatial 1-RDM has eigenvalues $(2, 1, 0, \ldots)$, and the von Neumann entropy of the normalized 1-RDM $S = -\sum_i (n_i/N) \log(n_i/N) = -\frac{2}{3}\log\frac{2}{3} - \frac{1}{3}\log\frac{1}{3} \approx 0.637$ nats is nonzero not from V_{ee} correlation but from the open-shell *antisymmetrization* of the single SD. The V_{ee} correlation contribution is the small differential $S_{\text{full}} - S_{\text{kin}} \approx 0.010$ nats at $Z = 3$. The proposition's S_A=0 statement applies in the *natural-orbital basis of h* where the open-shell SD has occupation $(1, 1, 1, 0, \ldots)$ in spin-orbital units, giving $S = 0$ trivially in that basis. In the spatial 1-RDM, the antisymmetrization entropy is the relevant floor, not zero.

Genuine H_1 degeneracy (Be, ^1S ground state at $n_{\max} = 3$). The bare graph H_1 has a 3-fold degenerate ground state from the unbroken $2s/2p$ shell degeneracy (the diagonal $-Z^2/2n^2$ does not split $2s$ from $2p$, and the κ-adjacency does not lift it in the $M_L = 0, M_S = 0$ sector). The H_1-only ground ensemble is mixed, the spatial 1-RDM has fractional occupations $(2, 0.76, 0.76, 0.47, 0, \ldots)$, and $S_{\text{kin}} = 1.23$ nats $\neq 0$. V_{ee} lifts the degeneracy and *reduces* entropy: $S_{\text{full}} = 0.902 < S_{\text{kin}}$. Here the proposition's premise fails outright; the floor is $S_{\text{kin}} > 0$, not zero.

The He two-electron case satisfies the proposition's premise in the strictest form: $1s^2$ closed shell exhausts the non-degenerate $n = 1$ shell, the GS is a single SD, the spatial 1-RDM has eigenvalues $(2, 0, \ldots)$, and the floor is $S_{\text{kin}} = 0$. Generically the proposition's spatial-basis floor $S_{\text{kin}} = 0$ applies to closed-shell ground states whose electron count exhausts an entire (n, l)-degenerate sub-shell at the h-eigenbasis level. For the GeoVac S^3 basis at $n_{\max} \geq 2$, this restricts the strict floor to He-like atoms and to systems whose closed shells fall entirely in the lowest $1s$ sub-shell. The proposition's operator content is general; its quantitative spatial-basis realization narrows.

Analytical degenerate perturbation theory for Be (EP-2N). The Be case admits a clean analytical treatment. At $n_{\max} = 2$, $M_L = M_S = 0$, the H_1 ground subspace is exactly 3-dimensional, spanned by $\{|1s^2 2s^2\rangle, |1s^2 2p_0^2\rangle, |1s^2\,(2p_{-1}2p_{+1})_{S=0}\rangle\}$ at common $E_{\text{kin}} = -5Z^2/4$. Diagonalizing V_{ee} within this subspace (a 3×3 degenerate-PT problem) at $Z = 4$ yields a ground state $0.98\,|2s^2\rangle - 0.14\,|2p_0^2\rangle - 0.14\,|(2p_{-1}2p_{+1})_{S=0}\rangle$ – 96% pure $2s^2$ with $\sim$ 2% admixture each of the two $2p$-pair configurations. The analytical spatial 1-RDM has diagonal occupations $(2.000, 1.924, 0.020, 0.037, 0.020)$ in the $(1s, 2s, 2p_{-1}, 2p_0, 2p_{+1})$ basis, giving $S_{\text{full}} = 0.794$ nats vs. the equal-mixture $S_{\text{kin}} = 1.358$ nats. V_{ee} thus acts as a *degenerate-PT lifter*: it selects the $2s^2$ configuration and reduces entropy by 42%. The full FCI calculation at $n_{\max} = 3$ (EP-2j) gives $S_{\text{full}} = 0.902$ in qualitative agreement, with the additional shells contributing modest extra correlation entropy on top of the analytical degenerate-PT prediction. This sharpens the post-degeneracy floor to a calculable quantity rather than the trivial $S_{\text{kin}} = 0$ of the non-degenerate case.

TABLE I. One-body vs. full ground-state entropy for He on the S^3 angular lattice. $S_{\rm full}$ and $S_{\rm kin}$ are the single-orbital von Neumann entropies of the FCI ground states of $H_{\rm full}$ and $H_{\rm kin}$, respectively. Data from `debug/data/ep1_entropy_partition.json` [12].

$n_{\max}$	$E_{\rm full}$ (Ha)	$S_{\rm full}$ (nats)	$S_{\rm kin}$ (nats)	$S_{\rm kin}/S_{\rm full}$
2	-2.8895	0.03955	-2.2×10^{-16}	-5.6×10^{-15}
3	-2.8931	0.04081	7.8×10^{-16}	1.9×10^{-14}

D. GeoVac specialization

For the GeoVac Coulomb lattice, $\mathcal{B}$ is the angular momentum eigenbasis $\{\phi_{nlm}\}$ on S^3. In this basis the one-body operator is

$$H_1 \;=\; \mathrm{diag}(\varepsilon_{nlm}) \;+\; \kappa\, L, \qquad \kappa = -\tfrac{1}{16}, \tag{3}$$

where $\varepsilon_{nlm} = -Z^2/(2n^2)$ is the hydrogenic spectrum and L is the GeoVac graph Laplacian. H_1 is non-degenerate at the closed-shell He ground state for all $n_{\max} \geq 1$. By the proposition, the single-orbital entropy of the H_1-only ground state vanishes in the natural orbital basis. In the angular momentum basis, H_1 is nearly but not exactly diagonal, and the natural orbitals deviate from $\{\phi_{nlm}\}$ by $O(|\kappa|/Z^2)$. The single-orbital entropy in $\mathcal{B}$ inherits an exponentially small projection contribution that is bounded above by the deviation of $\mathcal{B}$ from the natural-orbital basis of h.

III. NUMERICAL CONFIRMATION

We solve for the ground state of two He Hamiltonians on the S^3 angular momentum eigenbasis at $n_{\max} \in \{2, 3\}$:

- the full Hamiltonian $H_{\rm full} = H_1 + V_{ee}$, with V_{ee} from the exact-rational hypergeometric Slater integral evaluator [6, 12];
- the kinetic-only Hamiltonian $H_{\rm kin} = H_1$, identical to $H_{\rm full}$ with $V_{ee} \equiv 0$.

We diagonalize each in the $M_L = 0$ singlet sector by full configuration interaction, build the spatial 1-RDM, and report its von Neumann entropy in nats.

Table I reproduces the EP-1 data file [12].

The kinetic-only entropy is at the floating-point noise floor ($\sim 10^{-16}$ nats), confirming the proposition to machine precision. The full Hamiltonian's entropy ($S_{\rm full} \sim 4\times10^{-2}$ nats) is fourteen orders of magnitude larger. The natural-orbital occupations of the kinetic-only ground state at $n_{\max} = 3$ are $(2.000, 8.3\times10^{-22}, 0, 0, \ldots)$, exactly the $\{0, 1\}$ structure (here doubled by spin) predicted by the proposition. The full Hamiltonian's occupations $(1.988, 0.00896, 0.00093, 0.00093, \ldots)$ exhibit the $O(10^{-3})$ deviations from $\{0, 2\}$ characteristic of the two-body V_{ee} correlation.

The result is expected. Its value lies in making it visible on the concrete Coulomb lattice and in demonstrating that the floor is operator-theoretic, not basis-fitted.

IV. AREA LAW AS PAIR COUNTING

A. The Paper 5 statement

The framework synthesis paper [5] states (Section "Entropy and the second law", verbatim):

> The holographic entropy of shell n is $S_n = k \ln A_n + \text{const}$, where $A_n \propto n^4$ is the "surface area" (number of plaquettes).

The supporting microstate count was not given. The shell degeneracies in Paper 0 [4] are $g_n^{\rm orb} = n^2$ (orbital, no spin) and $g_n = 2n^2$ (with spin); cumulative state count is $N(n) = n(n+1)(2n+1)/3 \sim (2/3)n^3$. Neither yields n^4.

B. Pair counting on shell n

Define a *plaquette* on shell n as an ordered pair of sites,

$$A_n \;\equiv\; g_n^2 \;=\; 4\,n^4. \tag{4}$$

Equivalently, the unordered antisymmetrized pair count is

$$\tfrac{1}{2}\, g_n(g_n - 1) \;=\; n^2(2n^2 - 1) \;=\; 2\,n^4 + O(n^2), \tag{5}$$

and the adjacent-shell pair count is $g_n g_{n-1} = 4n^2(n-1)^2 = 4n^4 + O(n^3)$. All three two-body counts agree at leading order:

$$\log A_n \;=\; 4\,\log n \;+\; \text{const}. \tag{6}$$

A factor of four arises only at the two-body level: each electron contributes $\log g_n = 2\log n + \log 2$, and the two factors compose multiplicatively. At the one-body level, single-shell uniform entropy is $\log g_n = 2\log n + \log 2$ (off by a factor of two), and cumulative entropy $\log N(n) \sim 3\log n$ (off by a factor of 4/3). These are exhibited explicitly in the EP-3 analysis [15].

The area-law scaling $A_n \propto n^4$ asserted in Paper 5 is therefore correct, but it is a *two-body* statement. In the operator-theoretic language of Section II, the area law lives on the pair graph of the V_{ee} operator, not on the single-particle vertex set of H_1.

C. Reconciliation

Paper 5 is a predecessor framework synthesis with conjectural status. Its area-law claim is rigorously recovered here as a pair statement. Two consequences follow:

1. The area law cannot be derived from one-particle microstate counting on the GeoVac packing lattice. Any interpretation that assigns the n^4 scaling to a one-particle observable is mislabeled.
2. The information-theoretic content of the scaling is consistent with Section II: entropy on the GeoVac lattice is generated by the two-body operator, and its leading shell scaling counts pair states.

V. CUSP AS ENTROPY CONCENTRATION

The energy-graph characterization in v2.9.1 [16] constructs the singlet pair-state graph $\{|(ab)\rangle\}$ at $n_{\max} = 3$ ($N = 31$ nodes) and $n_{\max} = 4$ ($N = 101$ nodes), with edge weights $V_{IJ} = \langle (ab)|1/r_{12}|(cd)\rangle$ computed from the exact-float hypergeometric Slater integrals [6]. Two findings of that characterization bear directly on the operator-theoretic result of Section II.

Frobenius-mass diagonality. In the eigenbasis of the one-body operator H_1, V_{ee} is overwhelmingly diagonal:

$$\frac{\|\mathrm{diag}_{H_1} V_{ee}\|_F}{\|V_{ee}\|_F} = 0.920 \ (n_{\max} = 3), \ \ 0.892 \ (n_{\max} = 4), \tag{7}$$

using the unsquared Frobenius ratio (squared values are 0.847 and 0.795). The relative commutator $\|[H_1, V_{ee}]\|_F / \|H_1\|_F = 6.1\%$ at $n_{\max} = 3$, dropping only to 5.3% at $n_{\max} = 4$. The commutator does not vanish with basis size; H_1 and V_{ee} are genuinely non-commuting. The wavefunction-graph eigenbasis "almost" diagonalizes the entropy-generating operator, and the residual $\sim 10\%$ off-diagonal mass is precisely the cusp correlation that drives convergence from $\sim 3\%$ to 0.2% in graph-native CI.

Hot-node localization. On the same pair-state graph, the $(1s, 1s)$ vertex is simultaneously the maximum-diagonal entry ($V_{(1s,1s),(1s,1s)} = 5/8$, 42% of the next-largest diagonal), the largest weighted hub, and the head of the largest off-diagonal edge $(1s, 1s) \leftrightarrow (1s, 2s)$. The cusp does not distribute across pair-states; it concentrates on the unique pair-state where both electrons can simultaneously occupy small-r regions. Combined with Section II, this identifies the cusp not only as the energy-evaluation bottleneck [8] but as the locus of entanglement mass: the pair-state with maximal V_{ee} contribution is also the pair-state on which the two-body-generated entropy concentrates.

VI. EXCHANGE-CONSTANT TAXONOMY PLACEMENT

Paper 18 [8] classifies observables by the type of transcendental content their evaluation requires: *intrinsic* (rational graph-native quantities, e.g. Slater F^0 values), *calibration* (transcendental constants of the conformal projection, e.g. π from S^3 Weyl densities), *embedding* (constants that enter when the graph is mapped to a continuous metric, e.g. the Coulomb $1/r_{12}$ kernel), and *flow* (renormalization-style constants, e.g. Phillips–Kleinman).

Entropy fits the framework cleanly as a *projection-class observable*:

- On the bare combinatorial graph (vertex set $\{\phi_{nlm}\}$, one-body operator H_1 alone), the ground-state entropy is identically zero (Section II). No transcendental enters.
- Under the metric embedding of $1/r_{12}$ onto the pair-state graph, entropy becomes finite. Diagonal pair entries V_{ii} remain rational (Section V, [16]); spectral eigenvalues of V_{ee} on the pair graph are irrational, consistent with the embedding-class signature of $1/r_{12}$ in Paper 18.

Entropy is therefore an embedding-class observable in the Paper 18 taxonomy. The combinatorial graph is entropy-blind; entropy is the price of projecting onto the metric continuum.

This placement is consistent with the Paper 24 HO rigidity theorem [10]: the Bargmann–Segal harmonic oscillator lattice is bit-exactly π-free in rational arithmetic, and its one-body Hamiltonian has the same entanglement-inert character as H_1 here (it is one-body by construction). Whether two-fermion entanglement on the HO lattice carries the same area-law scaling as the Coulomb case is the subject of Section VII.

VII. TWO RIGIDITY RESULTS

This section states the two main entropy results beyond Section II's operator-theoretic floor. Section VII A establishes a rigidity theorem for the Bargmann–Segal harmonic-oscillator lattice: two-fermion closed-shell ground-state entropy is zero for any central two-body potential. Section VII B characterizes a universal two-variable scaling for Coulomb-lattice block entropy, asymptotic to second-order Rayleigh–Schrödinger perturbation theory. Both were originally phrased as speculative predictions; both are now established numerically, with Section VII A strengthened to a structural statement by Moshinsky–Talmi total-quanta conservation.

A. HO zero-entropy theorem for closed-shell two-fermion ground states

Outcome (EP-2b, this work). The prediction resolves cleanly: $S_{\mathrm{HO}} = 0$ *identically* at the ground state, not merely $\ll S_{\mathrm{Coulomb}}$. The mechanism is stronger than the "short-range kernel suppresses cusp concentration" heuristic originally stated: on the Bargmann–Segal HO basis, the one-body operator H_{HO} commutes exactly with *any* central two-body interaction $V(r_{12})$ because the

Moshinsky–Talmi transformation makes total HO quanta $N_1 + N_2 = N_{\rm rel} + N_{CM}$ a conserved quantum number. Consequently the ground state lies in the lowest-$N_{\rm tot}$ sector of H_{HO}; for a closed-shell system (two fermions in $M_L = 0$, $M_S = 0$), that sector is one-dimensional, and the ground state is a single Slater determinant $|(0s)^2\rangle$.

Numerical confirmation (EP-2b). With the Minnesota NN singlet-channel kernel at $\hbar\omega = 10\,\mathrm{MeV}$:

TABLE II. EP-2b: two-fermion HO on the Bargmann–Segal lattice, Minnesota $S = 0$ interaction. $E_{\rm full}$ is the FCI ground-state energy; $E_{\rm kin}$ is the noninteracting (HO-only) ground-state; $S_{\rm full}$ is the spatial 1-RDM von Neumann entropy of the interacting ground state. Data from `debug/data/ep2b_ho_two_fermion.json`.

$N_{\max}$	$E_{\rm full}$ (MeV)	$E_{\rm kin}$ (MeV)	$S_{\rm full}$ (nats)	$\lVert[H_{HO}, V]\rVert_F/\lVert H_{HO}\rVert_F$
2	22.185	30.000	0.0	7.9×10^{-16}
3	22.185	30.000	0.0	4.4×10^{-16}

The FCI ground state at both basis sizes is bit-identical to the single Slater determinant $|(0s)^2\rangle$; top-4 natural-orbital occupations are $(2.0, 0, 0, 0)$ exactly. The $N_{\max} = 2 \to 3$ comparison shows that enlarging the single- particle basis does *not* mix higher shells into the ground state: the commutator $[H_{HO}, V]$ is at the floating-point noise floor, so all eigenvectors of $H_{\rm full}$ are eigenvectors of H_{HO}, and the ground state is locked to the lowest-$N_{\rm tot}$ irreducible block.

Interpretation. The operator-theoretic statement of Section II should be read in its sharp form: entanglement is generated by the component of the two-body operator *off-diagonal in the H_1 eigenbasis restricted to the ground-state sector*, not merely by V's total Frobenius mass. On the HO lattice, V has nonzero total Frobenius mass (and even a $\sim 93\%$ diagonality ratio in the H_{HO} eigenbasis that parallels the Coulomb number), but its off-diagonal component between $N_{\rm tot}$ eigenspaces is *exactly zero*, so the ground state does not mix, and $S = 0$ exactly. The Coulomb kernel, in contrast, does not conserve any HO-like $N_{\rm tot}$ quantum number; it couples across shells (the 6–8% off-diagonal residue documented in Section V) and generates the $S \sim 4 \times 10^{-2}$ nats observed in EP-1.

Structural content. The result generalizes beyond Minnesota to *any* central two-body $V(r_{12})$ on the 3D isotropic HO basis: total-quanta conservation is a property of the Moshinsky–Talmi decomposition, not of the specific kernel. This is a discrete quantum-information reading of a standard nuclear-structure fact [14]. It provides a zero-entropy baseline for the Paper 18 exchange-constant taxonomy: on the HO graph, entropy is not merely "embedding-class with small amplitude" but vanishes identically at closed-shell ground states, consistent with the HO rigidity theorem of Paper 24 [10] (π-free graph certificate extends to π-free ground entropy).

Implementation. The experiment is driven by `debug/ep2b_ho_two_fermion.py` using the new `geovac.nuclear.ho_two_fermion` module (which composes the existing `geovac.nuclear.bargmann_graph`, `geovac.nuclear.moshinsky`, and `geovac.nuclear.minnesota` infrastructure from Paper 23 [9]). The numerical claims above are pinned by `tests/test_paper27_entropy.py::test_paper27_ep2b_ho_gs_is_s` and `::test_paper27_ep2b_ho_kinetic_interaction_commute`.

B. Universal scaling of Coulomb block entropy (verified, reframed)

Main result. Block ground-state entropy on the Coulomb lattice obeys a universal two-variable scaling in the dimensionless off-diagonal V_{ee} mass $\tilde{w}_B \equiv \lVert V_{ee}^{\rm off-diag(H_1)}\rVert_F/\lVert H_1\rVert_F$ and the dimensionless first H_1 gap $\delta_B \equiv \Delta E_1/\lVert H_1\rVert_F$:

$$S_B \;=\; A\,(\tilde{w}_B/\delta_B)^{\gamma}. \tag{8}$$

The local exponent approaches a basis-extension limit $\gamma_\infty \approx 1.96$ as both $Z \to \infty$ and $n_{\max} \to \infty$ (Richardson extrapolation on $n_{\max} = 2, 3, 4, 5$ local slopes at $Z = 15 \to 30$). This is *below* the non-degenerate second-order Rayleigh–Schrödinger prediction $\gamma = 2$. The persistent gap of ~ 0.04 at the basis limit is an aggregation effect: with more orbitals available, S_B accumulates contributions from many small admixtures rather than being dominated by the lowest excitation, and the aggregate sum scales with a slightly smaller exponent. Finite-window fits give $\gamma \in [2.1, 2.4]$, depending monotonically on both Z and $n_{\max}$. Across 22 calibration points (He-like $Z \in [2, 100]$ at $n_{\max} = 3$, plus a LiH bond-block R-sweep from 1.5 to 8 bohr) Eq. (8) fits at $R^2 = 0.990$ with $\gamma = 2.228$, $A = 0.157$; the only points with $> 25\%$ deviation are the dissociation-limit $R \in \{6, 8\}$ bohr LiH-bond points that saturate at $S_B = \log 2$ (the two-site ceiling).

Numerical confirmation. Three calibrations pin Eq. (8). (i) He-like isoelectronic sequence at $n_{\max} = 3$, $Z \in \{2, \ldots, 10\}$ gives $\gamma = 2.383$, $A = 8.16$, $R^2 = 0.998$ in the single-variable form $S_B = A\tilde{w}_B^{\gamma}$; this is equivalent to Eq. (8) because δ_B is essentially constant at 0.20 across the He-like family. The companion Z-scaling fit $S_B \sim Z^{\alpha_Z}$ gives $\alpha_Z = -2.613$, $R^2 = 0.995$, consistent with Paper 26's single-orbital entropy $\alpha_Z = -2.56$ [11]. (ii) Extending to $Z \in \{12, 15, 20, 30, 50, 100\}$ gives a monotonically decreasing local log-log slope: $\gamma_{\rm loc}(Z = 2) = 2.64 \to \gamma_{\rm loc}(Z = 100) = 1.92$, crossing 2 between $Z = 10$ and $Z = 12$. The underlying $S_B(\tilde{w}_B/\delta_B)$ curve is smooth, not a pure power law; local linearization over any finite window picks up a slope that depends on the window. (iii) Fixing the Z-window $Z \in \{2, \ldots, 30\}$ and sweeping $n_{\max} \in \{2, 3, 4\}$ gives local slopes at large Z of 1.992, 1.976, and 1.966 respectively: both Z and $n_{\max}$ drive γ toward 2 from above.

Methodological robustness. Three potential confounders were checked and did not affect the main result. (a) The apparent 40% prefactor offset between "core" and "lone-pair" blocks in an earlier 16-block

TABLE III. EP-2c pilot data: He-like isoelectronic sequence at $n_{\max} = 3$. Full data in `debug/data/ep2c_entropy_vs_vee_offdiag.json`. $\delta_B \approx 0.20$ (constant to 1%) across the entire table.

Z	S_B (nats)	$\tilde{w}_B$
2	4.08×10^{-2}	0.1052
3	1.12×10^{-2}	0.0648
4	4.98×10^{-3}	0.0470
6	1.82×10^{-3}	0.0306
8	9.48×10^{-4}	0.0226
10	5.86×10^{-4}	0.0179

TABLE IV. $\gamma(Z, n_{\max})$ surface (EP-2i + EP-2L). The local slope of S_B vs. $\tilde{w}_B/\delta_B$ between $Z = 15$ and $Z = 30$ decreases monotonically with $n_{\max}$, crossing 2 between $n_{\max} = 2$ and 3. Richardson extrapolation on $n_{\max} = 2, 3, 4, 5$ gives an asymptotic slope $\gamma_\infty \approx 1.96$, *below* the non-degenerate Rayleigh–Schrödinger value of 2. The global-window γ over $Z \in \{2, 3, 4, 6, 10, 15, 30\}$ is dominated by small-Z finite-basis residue and is consistently higher.

$n_{\max}$	γ (global, $Z \in [2, 30]$)	γ_{loc} ($Z : 15 \to 30$)
2	2.373	1.992
3	2.223	1.976
4	2.182	1.966
5	–	1.959
∞ (extrapolated)	—	≈ 1.96

multi-block fit is a pure Z-range artifact; repeating the same fit on the *identical* He-like family restricted to the core and lone-pair Z-sets reproduces the offset without any block-type variation. The underlying curve is not a pure power law, and log–log regression over different Z-windows picks up different local slopes (finite-basis residue at small Z inflates γ). (b) A bounded ansatz $S_B = \log 2 \cdot \tanh^2(c\,(\tilde{w}_B/\delta_B)^\gamma)$, motivated by the two-site $\log 2$ ceiling, fits markedly worse than the pure power law ($R^2 = 0.37$ vs. 0.90 on a 21-point mixed-geometry set) because the tanh Taylor expansion forces the small-argument slope to 2γ with a small prefactor c^2, incompatible with the $\sim$ 4-decade dynamic range in S_B. Three additional bounded ansatze (rational saturation, stretched exponential, Hill function) were tested in log-space on a combined 32-point dataset (EP-2g + EP-2k HF and H_2O bonds) and all give $R^2_{\log} \in [0.985, 0.989]$, statistically indistinguishable from the pure power law's 0.9895. The saturation at $\log 2$ is *discrete*, not smooth: only one or two LiH dissociation-limit points actually reach the ceiling, so saturation parameters cannot earn their fit cost. The pure power law Eq. (8) remains the best single functional form. (c) The $n_{\max} = 2, 3, 4$ sequence of global-window exponents $(\gamma, A) = (2.63, 0.73), (2.38, 0.18), (2.32, 0.09)$ shows monotone approach to $\gamma = 2$ from above, at fixed Z-window $\{2, 3, 4, 6, 10\}$; the local slope at the large-Z edge ($Z = 15 \to 30$) approaches $\gamma = 2$ more directly (Table IV).

Heavy-atom R-independence (EP-2k). HF and H_2O bond blocks were R-swept across $R \in [1.3, 3.0]$ bohr (HF) and $R_{OH} \in [1.5, 3.0]$ bohr (H_2O). Both are essentially *R-independent*: S_B varies by 0.4% for HF and 0.2% for H_2O across the entire range. At every R-point HF sits at $S_B/S_{\mathrm{predicted}} = 1.20 \pm 0.02$ above Eq. (8), and H_2O at 1.25 ± 0.01. In contrast, LiH varies by 70× across the same R-range and shows the saturation at $\log 2$ discussed above.

The mechanism is the Z asymmetry between center and partner. HF ($Z_F = 9$, $Z_H = 1$) and H_2O ($Z_O = 8$, $Z_H = 1$) have bonds dominated by the heavy atom; the H-side orbital contributes a perturbative correction whose amplitude depends on R only through cross-center V_{ne} matrix elements that vary slowly compared to the heavy-atom-dominated H_1 structure. LiH (Li-side $Z_{\mathrm{eff}} = 1$ = H-side Z) is the symmetric special case where bond stretching genuinely populates two equal-charge centers, driving the quasi-degenerate regime and the eventual $\log 2$ saturation. For chemically realistic heavy-atom bonds at or near equilibrium, the universal curve holds within a 25% band that is R-independent within $\sim$ 1%.

Bond-block qualifier. For bond blocks with two-center H_1, δ_B is the *essential* second variable: the LiH bond-block point at equilibrium ($\tilde{w}_B = 0.042$) would give $S_B \approx 4 \times 10^{-3}$ under the single-variable fit but the actual value is 0.079, a 20× deviation. Including δ_B collapses it onto Eq. (8) within the universal band. Bond blocks in isolation, however, are not a single well-defined object: the composed LiH bond block (within-block Hamiltonian only, Li-side $Z_{\mathrm{eff}} = 1$ = H-side Z) is *exactly degenerate* ($\delta_B = 0$) because the two sub-block 1s orbitals share the same effective charge. Only the cross-center V_{ne} terms introduced by the balanced builder [**?**] break the degeneracy and put the bond block at a finite $(\tilde{w}_B, \delta_B)$. A similar mechanism explains why BeH_2 at equilibrium saturates at $\log 2$ in the balanced builder: cross-block ERIs from the Be-core and the second Be–H bond compress δ_B from 0.204 (within-block) to 0.027 (balanced), an 8× reduction that crosses into the saturation regime. By contrast, the H_2O bond block shows no measurable shift ($S_{\mathrm{balanced}}/S_{\mathrm{composed}} = 1.002$), and its factor-of-2 residual from the universal-curve prediction is intrinsic. The operational statement is: Eq. (8) applies when one chooses a specific H_1 (including its cross-block contributions) and tracks both $\tilde{w}_B$ and δ_B for that choice.

Implementation. Drivers: `debug/ep2c_entropy_vs_vee_offdiag.py` (single-block pilot), `ep2c_multi_block.py` (multi-block), `ep2d_zrange_control.py` (Z-range control), `ep2f_full_library.py` (library + high-Z), `ep2e_bond_block.py` (bond-block divergence), `ep2g_two_variable_scaling.py` (universal w/δ scaling), `ep2h_universality_extension.py` (other bonds + $n_{\max}$ + bounded-form tests), `ep2i_deep_diagnostics.py` (BeH_2 cross-block decomposition, H_2O intrinsic offset, $\gamma(Z, n_{\max})$ surface), `ep2j_li_be_extension.py` (3e/4e atoms, non-degeneracy qualifier), `ep2k_heavy_atom_bond_sweep.py` (HF, H_2O bond R-sweeps), `ep2L_nmax5_gamma.py` ($n_{\max} = 5$ asymptote and Richardson extrapolation), `ep2M_two_scale_bounded.py` (rational, stretched-exp, Hill bounded-form fits, all negative), `ep2N_be_analytical.py` (Be 3×3 degenerate-PT diagonalization). Data under `debug/data/ep2{c,d,e,f,g,h,i}_*.json`; fits regression-locked by `tests/test_paper27_entropy.py::test_paper27_ep2c_dimensionless_power_law`, `::test_paper27_ep2c_multi_block_universality`, `::test_paper27_ep2d_z_range_artifact`, `::test_paper27_ep2g_universal_w_over_delta`, `::test_paper27_ep2h_nmax_residue_persists`, and `::test_paper27_ep2i_gamma_asymptote`.

C. Operational temperature decoding (Sprint TD Track 2)

The information-theoretic entropy of §II, $S_{\rm full} = S_{\rm vN}(\rho_1)$ of the bare-graph ground state, is the $T \to 0$ limit of the von Neumann entropy of the Gibbs spatial 1-RDM in standard quantum statistical mechanics. Sprint TD Track 2 (May 2026, `debug/sprint_td_track2_memo.md`) verifies this operationally on H, He, and Li^+ across 30 log-spaced temperatures spanning $0.005 \times \text{gap} \leq k_B T \leq 20 \times \text{gap}$, using the textbook Boltzmann-weighted ensemble over graph-native CI eigenstates.

The thermodynamic entropy decomposes cleanly as

$$S_{\rm thermo}(\beta) = k_B \cdot S_{\rm microstate\,info}(\beta), \tag{9}$$

where the right-hand side is the Shannon entropy of the microstate-resolved Boltzmann distribution. The identity holds at machine precision (max residual 4.8×10^{-14} across all $3 \times 30 = 90$ test points). This is standard quantum statistical mechanics; the sprint's content is the explicit verification on graph-native CI states.

The von Neumann entropy of the Gibbs *spatial 1-RDM*, $S_{\rm vN}(\rho_1(\beta))$ with $\rho_1(\beta) = \sum_n p_n(\beta)\,\rho_1^{(n)}$, is the "decoded-temperature spatial residual" — what remains of the information content after the apparatus parameter β is factored out. At $T \to 0$ this residual equals $S_{\rm full}$(GS) of §II bit-identically:

TABLE V. Sprint TD Track 2: $T \to 0$ residual matches $S_{\rm full}$(GS) of §II. Data `debug/data/sprint_td_track2.json`.

System	$S_{\rm vN}(\rho_1(\beta \to \infty))$	$\lvert\Delta\rvert$ vs. $S_{\rm full}$(GS)
H ($Z = 1$, $n_{\max} = 6$)	0.000000	0 (single det. rigidity)
He ($Z = 2$, $n_{\max} = 3$)	0.040811	3.1×10^{-10}
Li^+ ($Z = 3$, $n_{\max} = 3$)	0.011212	3.2×10^{-10}

In the $T \to \infty$ limit the residual approaches $\log(n_{\rm orbitals}) = \log 14 \approx 2.639$ for both $N_e = 2$ systems, the standard Boltzmann thermal-mixing limit over the graph basis. Lindblad concavity bounds hold at every β ($\max_n S_n \leq S_{\rm vN}(\rho_1(\beta)) \leq S_{\rm apparatus} + \sum_n p_n S_n$).

Operationally, this positions the Paper 27 information entropy as the dimensionless content of the GeoVac graph that remains after a finite-temperature observation projection (Paper 34 §III.15) is undone. Sprint TD Track 5 separately PSLQ-tested whether $S_{\rm full}$(GS) values for He, Li^+, and He at $n_{\max} = 4$ sit in the master Mellin engine ring (Paper 32 §VIII case-exhaustion theorem and Paper 18 §III.7); the test returned a clean negative against a 12,312-term canonical basis at 150 dps across all three systems, with control-class baseline matching to $z = 0$. Six orders of magnitude in coefficient relaxation ($10^3 \to 10^6$) does not change the verdict. GeoVac correlation entropy is genuinely von Neumann-only; it is not a master Mellin output. This is structurally distinct from spectral-action entropies (e.g., the Connes–Chamseddine area term on Euclidean Schwarzschild, which lives in M_2). $S_{\rm full}$ joins $K = \pi(B + F - \Delta)$, the Sprint MR-C L_2 next-order constant $c \approx 4.10932\ldots$, and the Wolfenstein parameters of Sprint W3 on the list of natural framework constants outside the master Mellin ring.

We do not speculate about gravitational, holographic, or black-hole entropy in this paper. Whether the pair-counting re-interpretation of the area law in Section IV admits a continuum geometric reading is a separate question; the operator-theoretic content stands without it.

VIII. CONCLUSION

Five results have been established on GeoVac lattices for closed-shell two-fermion ground states. First, the von Neumann entropy of any single-particle reduced density matrix derived from the *non-degenerate* ground state of the one-body Hamiltonian alone is identically zero; numerically, $S_{\rm kin}/S_{\rm full} \sim 10^{-14}$ on He at $n_{\max} = 2, 3$. All ground-state entanglement is generated by the two-body operator V_{ee}. Second, the area-law scaling $S_n \propto 4 \log n$ asserted on one-particle grounds in Paper 5 [5] is recovered as a two-body pair-counting statement $A_n = g_n^2 \propto n^4$, with the factor of four as the two-body signature. Third, the cusp localizes as a single hot node on the V_{ee} pair graph, making the loci of energy correlation, entropy generation, and cusp mass coincident. Fourth, on the Bargmann–Segal harmonic oscillator lattice, two-fermion closed-shell ground-state entropy is identically zero for *any* central two-body potential, a rigidity statement structurally parallel to the Fock and Bargmann–Segal theorems of Paper 24. Fifth, on the Coulomb lattice, block entropy obeys a universal two-variable scaling $S_B = A\,(\tilde{w}_B/\delta_B)^\gamma$ in the dimensionless V_{ee} off-diagonal mass and the first H_1 gap, with the local exponent extrapolating to $\gamma_\infty \approx 1.96$ as $Z, n_{\max} \to \infty$ – below the naive second-order Rayleigh–Schrödinger value of 2, by a residue attributed to multi-shell aggregation in the basis-extension limit.

These results place entropy in the exchange-constant taxonomy of Paper 18 [8] as a projection-class observable: zero on the combinatorial graph, finite only after the metric embedding of $1/r_{12}$. The two-body character of the area-law scaling is structural, not incidental; the predicted contrast between Coulomb and harmonic oscillator two-fermion lattices is sharp and falsifiable.

EQUATION VERIFICATION

Per Section 13.4a of the project guidelines, every numerical claim in this paper is traceable to a debug data file or to a test in the codebase.

- Table I: `debug/data/ep1_entropy_partition.json`. **Verified** by `tests/test_paper27_entropy.py::test_paper27_table1_ep1_reproduction`, which re-runs the EP-1 pipeline (FCI of H_1 vs. $H_1 + V_{ee}$ on the He angular momentum basis, via `geovac/casimir_ci.py` and `geovac/hypergeometric_slater.py`) and checks $|S_{\rm kin}| < 10^{-12}$ and $|S_{\rm kin}|/S_{\rm full} < 10^{-10}$ at $n_{\max} \in \{2, 3\}$.
- Eq. (4)–(6): combinatorial identities on Paper 0 shell degeneracies $g_n = 2n^2$. **Verified** symbolically by `tests/test_paper27_entropy.py::test_paper27_area_law_pair_counts` and `::test_paper27_area_law_leading_order_agrees`, with companion test `::test_paper27_one_body_counts_differ_from_area_law` confirming the one-particle slope is 2 (single-shell) or 3 (cumulative), distinct from the two-body slope 4. Hand-tabulation in EP-3 analysis [15].
- Section V numerical values ($\|V_{ee}^{\mathrm{diag}\,(H_1)}\|_F/\|V_{ee}\|_F$, relative commutator, hot-node weights): produced by `debug/energy_graph_exploration.py`, data in `debug/data/energy_graph_nmax{3,4}.json`. **Verified** by `tests/test_paper27_entropy.py::test_paper27_energy_graph_structural_invariants` (commutator ratio, diagonal fraction, and exact $V_{(1s,1s),(1s,1s)} = 5/8$ at both $n_{\max} = 3$ and 4); non-vanishing of the commutator in the $n_{\max} = 3 \to 4$ extension is pinned by `::test_paper27_commutator_saturates_not_vanishes` (`--slow`).
- Table III and Eq. (8) (the dimensionless power law $S_B = 8.16\,\tilde{w}_B^{2.383}$, $R^2 = 0.998$): **Verified** by `tests/test_paper27_entropy.py::test_paper27_ep2c_dimensionless_power_law`, data in `debug/data/ep2c_entropy_vs_vee_offdiag.json`.
- Table II and Section VII A numerical claims ($S_{\rm HO} = 0$, occupation $(2.0, 0, 0, 0)$, commutator at floating-point noise): **Verified** by `tests/test_paper27_entropy.py::test_paper27_ep2b_ho_gs_is_single_determinant` and `::test_paper27_ep2b_ho_kinetic_interaction_commute`, data in `debug/data/ep2b_ho_two_fermion.json`.
- Section VI embedding-class placement: diagonal pair-state V_{ii} rationality and spectral irrationality verified in EP-3 / energy-graph analysis [16]; consistent with Paper 18 classification.

The operator-theoretic statement of Section II is a known free-fermion fact [1]; no new verification is required beyond the numerical confirmation in Section III.

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